

* Applications of Curve Tracing:-

Knowledge of tracing of curves is useful in applications of integrations in finding area, mass C.G. etc. These applications involving integrations need approximate limits. Therefore, it becomes necessary to trace the curve and then the limits can be obtained from the shape of the curve.

* Rules to trace the Cartesian curves:-

* Rule 1: Symmetry

(A) About X-axis :- IF the given equation has even powers of y everywhere, then curve is symmetrical about X-axis.

ex) $x = y^2$

(B) About Y-axis :- IF the given equation has even powers of x everywhere, then curve is symmetrical about Y-axis.

Ex) $y = x^2$

(C) Symmetry about the line $y = x$ (Implicit Relations)

Replace x by y and y by x , if the given eqn remains unchanged then curve is symmetric about the line $y = x$.

Ex) $xy = c$

(D) Symmetry about the line $y = -x$ (Implicit Relations)

Replace x by $-y$ and y by $-x$, if the given equation remains unchanged then curve is symmetrical about the line $y = -x$.

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(E) Symmetric in opposite quadrants:-

(For Implicit Relations only)

Replace x by $-x$ and y by $-y$, if equation remains unchanged then curve is symmetrical in opposite quadrant.

Ex) $y = x^3$

* **Implicit Equations:-** The equations in which we can not take terms of x on one side and terms of y on other side.

Ex) $x^2 + 2xy + y^2 = 0$

* **Explicit Equations:-** The equations in which we can take terms of x on one side and terms of y on other side.

$$y^2 = \frac{2a - x}{x^3}$$

* **Rule 2: Origin**

(A) Origin:- put $x = 0$ if we get $y = 0$ then curve passes through origin or vice versa.

(B) Tangents at origin:-

If the curve passes through the origin then to find the tangent at origin, Equate the lowest degree terms taken together to zero.
i.e.

$$\text{Lowest degree terms} = 0$$

* **Rule 3: Intersection with co-ordinate axes**

(A) Intersection with X-axis:- put $y = 0$ and find different values of x .

(B) Intersection with Y-axis: - put $x=0$ and find different values of y .

(C) Intersection with the line $y=x$ or $y=-x$
(For Implicit Relations only)

(1) IF the curve is symmetric about the line $y=x$ then find the points of intersection by putting $y=x$ in given equation.

(2) IF the curve is symmetric about the line $y=-x$ then find the points of intersection by putting $y=-x$ in given equation.

* Note: - Points given in above Rule 3 are called as special points.

* Rule 4: Tangents at special points:

To find tangents at special points, first find $\frac{dy}{dx}$.

(i) $\left(\frac{dy}{dx}\right)_{(x_1, y_1)} = 0$ then tangent is parallel to X-axis.

(ii) $\left(\frac{dy}{dx}\right)_{(x_1, y_1)} = \infty$, then tangent is parallel to Y-axis.

(iii) $\left(\frac{dy}{dx}\right)_{(x_1, y_1)} = +ve$ then tangent makes acute angle with X-axis.

(Implicit Relations only)

(iv) $\left(\frac{dy}{dx}\right)_{(x_1, y_1)} = -ve$ then tangent makes obtuse angle with Y-axis.

(Implicit Relations only)

* Rule 5:- Asymptotes

Asymptotes are tangents to the curve at infinity.

(A) Asymptotes parallel to X-axis:-

Asymptotes parallel to X-axis can be obtained by equating to zero, coefficient of highest power of x taken together.
i.e.

coefficient of highest power of $x = 0$
(All terms)

(B) Asymptotes parallel to Y-axis:-

Asymptotes parallel to Y-axis can be obtained by equating to zero, coefficient of highest power of y taken together.
i.e.

coefficient of highest power of $y = 0$
(All terms)

(C) Oblique Asymptotes:- (For Implicit Relations only)

Let $y = mx + c$ be the oblique asymptote
put $y = mx + c$ in the given eq and by comparing L.H.S. and R.H.S. find the values of m and c .

* Rule 6: Region of absence:-

Arrange the given eqn as $y^2 = f(x)$
or $x^2 = f(y)$ then check for negative values of

- (i) odd powers of x (or y)
- (ii) Addition
- (iii) Subtraction.

$\therefore$ Any value is negative if it is less than 0.

ex) $x + a < 0$

$$x < -a.$$

curve does not exist for values less than $-a$.

Que: Trace the curve $y^2(2a-x) = x^3$.

(Explicit Relation)

$\Rightarrow$ * Rule 1 Symmetry: - Even powers of y everywhere, therefore curve is symmetrical about X-axis.

* Rule 2 (A) Origin: -

Given $y^2(2a-x) = x^3$

put $x=0$

$$y^2(2a-0) = 0^3$$

$$y^2(2a) = 0$$

$$y^2 = \frac{0}{2a}$$

$$y^2 = 0$$

$$y = 0$$

$\therefore$ when $x=0$ we get $y=0$ i.e. $(0,0)$

Curve passes through the origin.

(B) Tangents at origin: -

(Arrange the eqn equal to 0 without any bracket multiplication).

Given $y^2(2a-x) = x^3$

$$2ay^2 - xy^2 = x^3$$

$$2ay^2 - xy^2 - x^3 = 0$$

$$\downarrow$$

2

$$\downarrow$$

3

$$\downarrow$$

3

$\longrightarrow$ powers of x and y

To find tangents at origin, equate to zero lowest degree term taken together.

i.e.

lowest degree terms = 0

$$2ay^2 = 0$$

$$y^2 = 0$$

$$y = 0$$

As tangent is a line and $y = 0$ represent a line which is x -axis.

$\therefore x$ -axis is tangent at origin.

* Rule 3: Intersection with axes

(A) Intersection with x -axis: -

$$\text{As } y^2(2a-x) = x^3$$

$$\text{put } y = 0$$

$$0^2(2a-x) = x^3$$

$$0 = x^3$$

$$x = 0$$

i.e. $(0,0) \longrightarrow$ origin

(B) Intersection with y -axis: -

$$y^2(2a-x) = x^3$$

$$\text{put } x = 0$$

$$y^2(2a-0) = 0^3$$

$$2ay^2 = 0$$

$$y^2 = 0$$

$$y = 0$$

i.e. $(0,0) \longrightarrow$ origin

*

Rule 4: Tangents at special points:.

We got $(0,0)$ and $(0,0)$ but it is not a special point. Hence there will be no

tangents at special points.

* Rule 5: Asymptotes

(Arrange the eqn equal to zero without any bracket multiplication)

$$\text{i.e. } y^2(2a-x) = x^3$$

$$2ay^2 - xy^2 = x^3$$

$$2ay^2 - xy^2 - x^3 = 0$$

(A) Asymptote parallel to X-axis :-

$$2ay^2 - xy^2 - x^3 = 0$$

↓

↓

↓

0

1

3

→ powers of x only

coefficient of highest power of x = 0

i.e.

$$-1 = 0$$

which is not possible.

∴ No asymptotes parallel to X-axis

(B) Asymptote parallel to Y-axis :-

$$2ay^2 - xy^2 - x^3 = 0$$

↓

↓

↓

2

2

0

→ powers of y only.

To find asymptote parallel to Y-axis, equate to zero, the coefficient of highest power of y taken together.

i.e. coefficient of highest power of y = 0

$$(2a - x) = 0$$

$$2a = x$$

$$x = 2a$$

$x = 2a \longrightarrow$ line / tangent / Asymptote
 $\therefore x = 2a$ is the asymptote parallel to Y-axis

* Rule 6 :- Region of absence :-

As $y^2(2a-x) = x^3$

$$y^2 = \frac{x^3}{(2a-x)}$$

We check negative values given by

(i) odd powers of x

$$x^3 < 0$$

$$\therefore \boxed{x < 0}$$

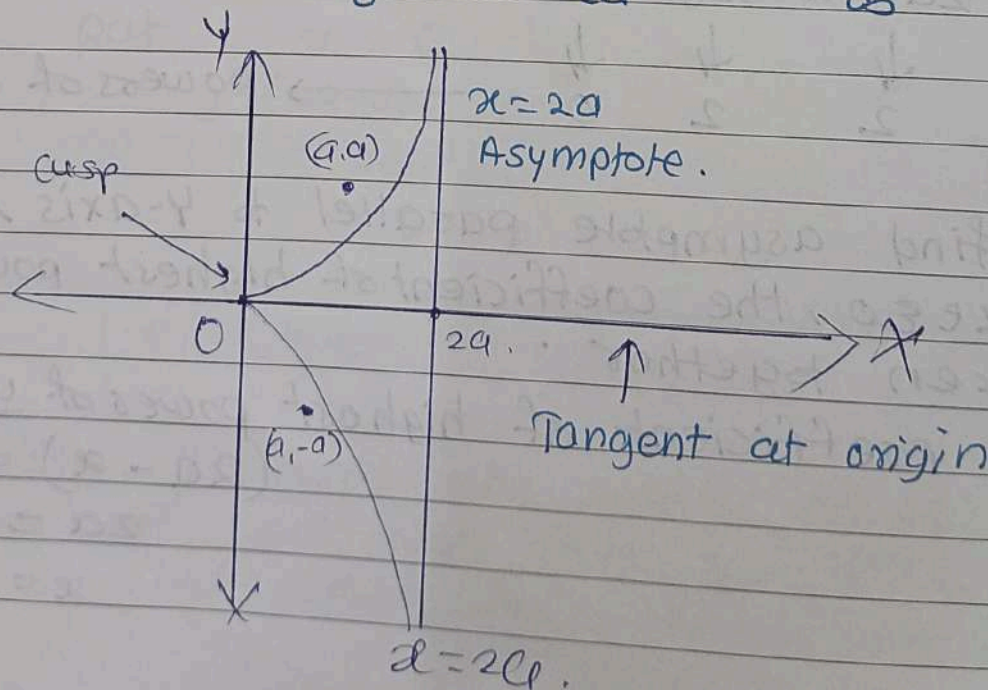
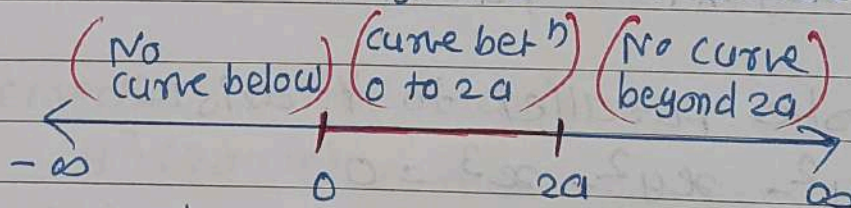
(ii) Subtraction

$$2a - x < 0$$

$$2a < x$$

$$\text{i.e. } x > 2a$$

The scale of Region of absence is as follows:



Que) Trace the curve $xy^2 = a^2(a-x)$

⇒ Rule 1: Symmetry

Given $xy^2 = a^2(a-x)$

Even powers of y everywhere, therefore curve is symmetrical about X-axis.

* Rule 2: (A) Origin

As $xy^2 = a^2(a-x)$

put $x=0$

$$0y^2 = a^2(a-0)$$

$$0 = a^3$$

$$a^3 = 0$$

i.e. when we put $x=0$, we do not get $y=0$

i.e. Curve does not pass through origin.

(B) Tangents at origin: As curve does not pass through the origin, there will not be any tangents at origin.

* Intersection with axes:-

As $xy^2 = a^2(a-x)$

put $y=0$

$$\therefore 0 = a^2(a-x)$$

$$\frac{0}{a^2} = a-x$$

$$x = a$$

∴ when $y=0$ we get $x=a$, i.e. $(a,0)$.

(This is a point not a line since we are finding the intersection).

∴ $(a,0)$ is a special point.

* Intersection with Y-axis

As $xy^2 = a^2(a-x)$

put $x=0$

$$0y^2 = a^2(a-0)$$

$$0 = a^3$$

$$a^3 = 0$$

when we put $x=0$, we are not getting any value of y . Therefore there is no intersection with Y-axis

* Rule 4: Tangents at special points:-

(Arrange the given eqⁿ as equal to zero)

$$xy^2 = a^2(a-x)$$

$$xy^2 = a^3 - a^2x$$

$$xy^2 - a^3 + a^2x = 0$$

Differentiating w.r.t. x

$$\frac{d}{dx}(xy^2) - \frac{d}{dx}(a^3) + \frac{d}{dx}(a^2x) = 0$$

$$\left(\because \frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx} \right)$$

$$\therefore x \frac{d}{dx}(y^2) + y^2 \frac{d}{dx}(x) - 0 + a^2 \frac{d}{dx}(x) = 0$$

$$x(2y) \frac{dy}{dx} + y^2(1) - 0 + a^2(1) = 0$$

$$2xy \frac{dy}{dx} = -y^2 - a^2$$

$$\frac{dy}{dx} = \frac{-y^2 - a^2}{2xy}$$

$\therefore \frac{dy}{dx}$ at special point $(a, 0)$

put $x=a, y=0$

$$\left(\frac{dy}{dx}\right)_{(a,0)} = \frac{-0-a^2}{2(a)(0)} = \frac{-a^2}{0} = \infty \Rightarrow \text{parallel to Y-axis.}$$

$\therefore$ Hence at special point $(a,0)$ tangent is parallel to Y-axis.

* Rule 5: Asymptotes

(A) Asymptotes parallel to X-axis:-
 $xy^2 = a^2(a-x)$

$$xy^2 - a^3 + a^2x = 0$$

$$\begin{array}{ccc} \downarrow & \downarrow & \downarrow \\ 1 & 0 & 1 \end{array}$$

$\longrightarrow$ powers of x .

To find the asymptotes parallel to X-axis.

Let

coefficient of highest power of x taken together $= 0$

$$y^2 + a^2 = 0$$

$$y^2 = -a^2$$

which is not possible as $y = \sqrt{-a^2}$ will lead to an imaginary number.

$\therefore$ No asymptote parallel to X-axis.

(B) Asymptote parallel to Y-axis:-

$$xy^2 - a^3 + a^2x = 0$$

$$\begin{array}{ccc} \downarrow & \downarrow & \downarrow \\ 1 & 0 & 0 \end{array}$$

$\longrightarrow$ powers of y

$\therefore$ coefficient of highest power of $y = 0$

$x = 0 \rightarrow$ line.

$x=0 \Rightarrow y$ -axis
So y -axis is asymptote parallel to y -axis

* Rule 6: Region of Absence:-

As $xy^2 = a^2(a-x)$

$$y^2 = \frac{a^2(a-x)}{x}$$

then check negative values of

(i) odd powers of x
i.e. $x < 0$

(ii) subtraction

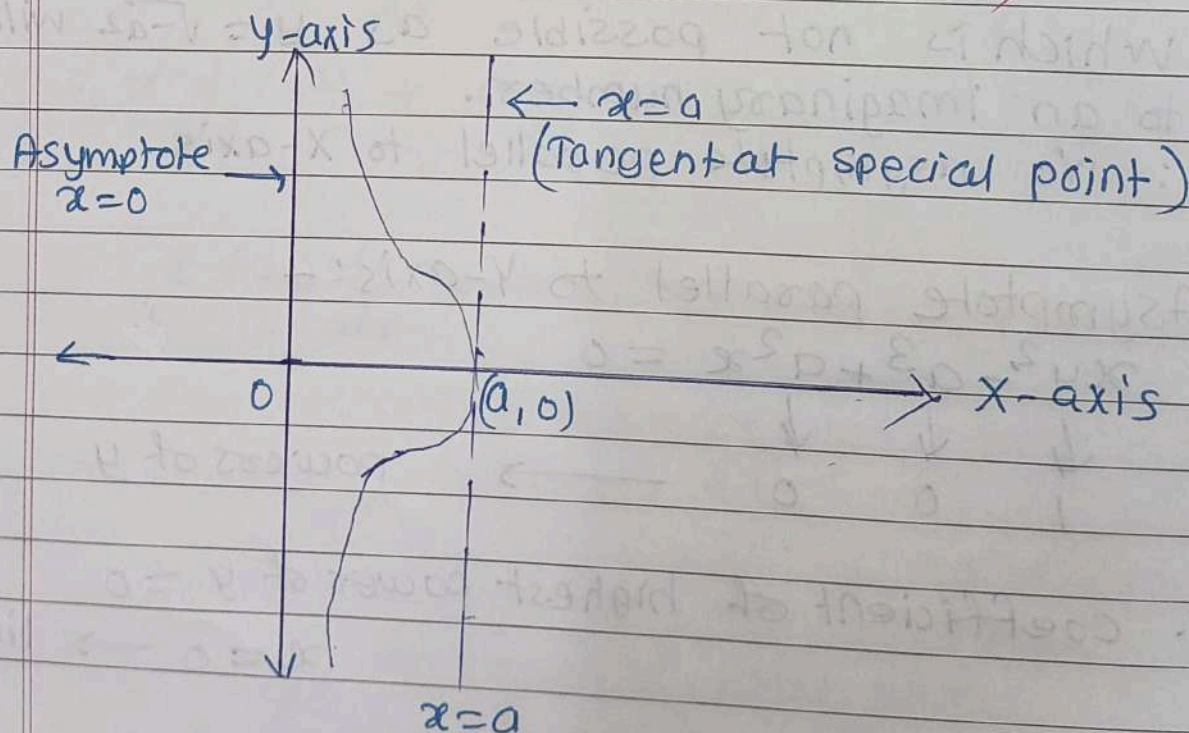
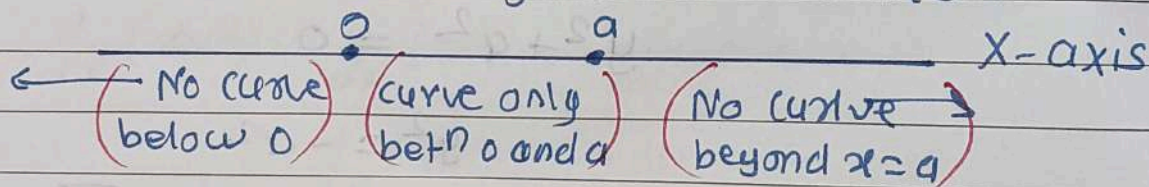
$$a-x < 0$$

$$a < x$$

i.e. $x > a$

i.e. there is no curve in the region $x < 0$
and $x > a$.

The scale of region of absence.



que) Trace the curve $y^2 = x^2(1-x)$

Rule 1: Symmetry

As $y^2 = x^2(1-x)$

$\therefore$ Even powers of y everywhere, curve is symmetrical about x -axis.

* Rule 2: (A) Origin

As $y^2 = x^2(1-x)$

put $x=0$

$$y^2 = 0^2(1-0)$$

$$y^2 = 0$$

$$\therefore y = 0$$

i.e. for $x=0$ we get $y=0$, i.e. $(0,0)$.

Hence curve passes through the origin.

(B) Tangents at origin

$$\therefore y^2 = x^2(1-x)$$

$$y^2 = x^2 - x^3$$

$$y^2 - x^2 + x^3 = 0$$

$$\therefore \downarrow \quad \downarrow \quad \downarrow$$

$$2 \quad 2 \quad 3$$

$\longrightarrow$ powers of x and y .

To find tangents at origin

lowest degree terms $= 0$

taken together

$$y^2 - x^2 = 0$$

$$y^2 = x^2$$

$$\text{i.e. } y = \pm x$$

The lines $y = x$ and $y = -x$ are tangents at origin.

* Rule 3: Intersection with axes

(A) Intersection with X-axis :-

$$\text{Given, } y^2 = x^2(1-x)$$

$$\text{put } y=0$$

$$0^2 = x^2(1-x)$$

$$x^2(1-x) = 0$$

$$x^2 = 0$$

$$\text{or } 1-x = 0$$

$$x = 0$$

$$\text{or } 1 = x$$

$$\boxed{x=0}$$

$$\text{or}$$

$$\boxed{x=1}$$

$$\text{When } y=0 \Rightarrow x=0, 1.$$

$\therefore (0,0)$ and $(1,0)$ are the points of intersection with X-axis. Out of which $(1,0)$ is the special point.

(B) Intersection with Y-axis :-

$$\text{Given } y^2 = x^2(1-x)$$

$$\text{put } x=0$$

$$y^2 = 0^2(1-0)$$

$$y^2 = 0$$

$$y = 0$$

for $x=0$ we get $y=0$ i.e. $(0,0)$ - origin.

* Rule 4: Tangents at special point

$$\text{As } y^2 = x^2(1-x)$$

$$y^2 = x^2 - x^3$$

$$y^2 - x^2 + x^3 = 0$$

diff w.r.t. x

$$\frac{d}{dx}(y^2) - \frac{d}{dx}(x^2) + \frac{d}{dx}(x^3) = 0$$

$$2y \frac{dy}{dx} - 2x + 3x^2 = 0$$

$$2y \frac{dy}{dx} = 2x - 3x^2$$

$$\frac{dy}{dx} = \frac{2x - 3x^2}{2y}$$

point $(1,0) \Rightarrow x=1$ and $y=0$

$$\therefore \left(\frac{dy}{dx} \right)_{(1,0)} = \frac{2(1) - 3(1)^2}{0} = \frac{-1}{0} = \infty \Rightarrow \text{Parallel to Y-axis}$$

At special point $(1,0)$, tangent is parallel to Y-axis.

* Rule 5: Asymptotes

(A) Asymptotes parallel to X-axis

$$\text{As } y^2 - x^2 + x^3 = 0$$

$$\begin{array}{ccc} \downarrow & \downarrow & \downarrow \\ 0 & 2 & 3 \end{array} \longrightarrow \text{powers of } x.$$

To find asymptotes parallel to x-axis

Coefficient of highest power of x taken together = 0

$$1 = 0$$

which is not possible.

$\therefore$ No asymptotes parallel to X-axis

(B) Asymptotes Parallel to Y-axis

$$\text{We have } y^2 - x^2 + x^3 = 0$$

$$\begin{array}{ccc} \downarrow & \downarrow & \downarrow \\ 2 & 0 & 0 \end{array} \longrightarrow \text{powers of } y.$$

$\therefore$ To find asymptote parallel to Y-axis

let coefficient of highest power of y = 0

i.e. $1=0$
which is not possible.
 $\therefore$ No Asymptote parallel to Y-axis.

* Rule 6: Region of Absence:—

We have $y^2 = x^2(1-x)$

$$y^2 = x^2(1-x)$$

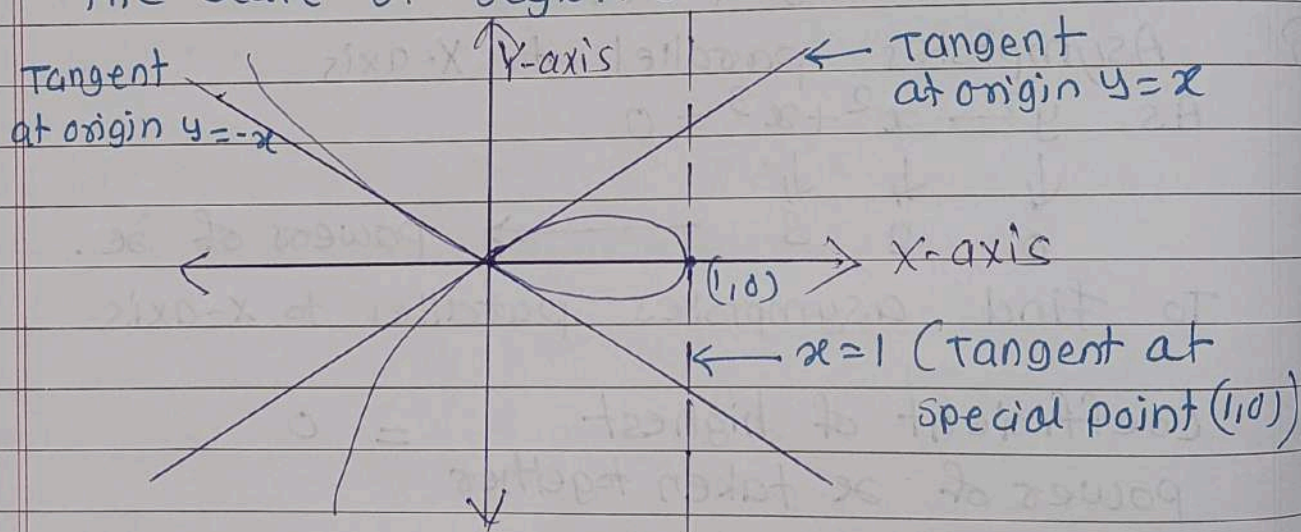
check the negative values of
(i) subtraction

$$\text{i.e. } 1-x < 0$$

$$1 < x$$

$$x > 1$$

The scale of region of absence is



HW Que) Trace the curve $y^2(a^2-x^2) = a^3x$

$\Rightarrow$ Symmetric about X-axis, Passes through origin and $x=0$ i.e. Y-axis is tangent at origin.

Asymptotes: $y=0, x=a, x=-a$.

Region of Absence: $x < 0, a < x, x > (a)$.

* Example on Implicit Relations:-

Que:- Trace the curve $x^3 + y^3 = 3axy \Rightarrow$ Implicit Relation.

Rule 1: Symmetry

(A) About X-axis:-

No even powers of y everywhere, therefore curve is not symmetrical about X-axis.

(B) About Y-axis:-

No even powers of x , therefore curve is not symmetrical about Y-axis.

(C) About the line $y = x$.

Given $x^3 + y^3 = 3axy$

Replace x by y and y by x .

$$\therefore y^3 + x^3 = 3ayx$$

$$\text{i.e. } x^3 + y^3 = 3axy$$

$\therefore$ Equation remains unchanged.

$\therefore$ Curve is symmetrical about the line $y = x$.

* Rule 2: (A) origin

We have $x^3 + y^3 = 3axy$

$$\text{put } y = 0$$

$$x^3 + 0^3 = 3ax(0)$$

$$x^3 = 0$$

$$\Rightarrow x = 0$$

i.e. for $y = 0$ we get $x = 0$, i.e. $(0, 0)$

$\therefore$ Curve passes through the origin.

(B) Tangents at origin:-

Given: $x^3 + y^3 = 3axy$

$$x^3 + y^3 - 3axy = 0$$

$$\frac{1}{3} \quad \frac{1}{3} \quad \frac{1}{2}$$

—————> powers of x and y

To find tangents at origin,
lowest degree term = 0

$$-3axy = 0$$

$$xy = \frac{0}{-3a}$$

$$xy = 0$$

$$\therefore x = 0 \text{ or } y = 0$$

↓

Y-axis

↓

X-axis

$\therefore$ X-axis and Y-axis are tangents at origin

* Rule 3 : Intersection with axes

(A) Intersection with X-axis:-

$$\text{Given } x^3 + y^3 = 3axy$$

$$\text{put } y = 0$$

$$x^3 + 0^3 = 3ax(0)$$

$$x^3 = 0$$

$$x = 0$$

$$\therefore (0, 0) \Rightarrow \text{origin}$$

(B) Intersection with Y-axis:-

$$\text{As } x^3 + y^3 = 3axy$$

$$\text{put } x = 0$$

$$0^3 + y^3 = 3a(0)y$$

$$y^3 = 0$$

$$\therefore \boxed{y = 0}$$

$$\text{Again } (0, 0) \Rightarrow \text{origin.}$$

(C) Intersection with the line $y = x$

(Only for Implicit Equations)

$$\Rightarrow \text{As } x^3 + y^3 = 3axy$$

$$\text{put } x = y$$

$$y^3 + y^3 = 3a(y)(y)$$

$$2y^3 = 3ay^2$$

$$2y^3 - 3ay^2 = 0$$

$$y^2(2y - 3a) = 0$$

$$y^2 = 0 \text{ or } 2y - 3a = 0$$

$$y = 0 \text{ or } y = \frac{3a}{2}$$

when $y = 0$, we have $x = y \Rightarrow x = 0$.

(A-1) i.e. $(0, 0) \Rightarrow$ origin

when $y = \frac{3a}{2}$, we have $x = y = \frac{3a}{2}$ $\left(\frac{3a}{2}, \frac{3a}{2}\right)$.

$\left(\frac{3a}{2}, \frac{3a}{2}\right)$ is a special point.

* Rule 4: Tangents at special points.

we have $x^3 + y^3 - 3axy = 0$

Diff. w.r.t. x

$$\frac{d}{dx}(x^3) + \frac{d}{dx}(y^3) - \frac{d}{dx}(3axy) = 0$$

$$\therefore 3x^2 + 3y^2 \frac{dy}{dx} - 3a \left[x \frac{dy}{dx} + y(1) \right] = 0$$

$$3x^2 + 3y^2 \frac{dy}{dx} - 3ax \frac{dy}{dx} - 3ay = 0$$

$$(3y^2 - 3ax) \frac{dy}{dx} + 3x^2 - 3ay = 0$$

$$(3y^2 - 3ax) \frac{dy}{dx} = 3ay - 3x^2$$

$$\boxed{\frac{dy}{dx} = \frac{3ay - 3x^2}{3y^2 - 3ax}}$$

$$\left(\frac{3a}{2}, \frac{3a}{2}\right) \Rightarrow x = y = \frac{3a}{2}$$

$$\begin{aligned}
 \therefore \left(\frac{dy}{dx} \right)_{\left(\frac{3a}{2}, \frac{3a}{2} \right)} &= \frac{a \left(\frac{3a}{2} \right) - \left(\frac{3a}{2} \right)^2}{\left(\frac{3a}{2} \right)^2 - a \left(\frac{3a}{2} \right)} \\
 &= \frac{\left(\frac{3a^2}{2} - \frac{9a^2}{2} \right)}{\left(\frac{9a^2}{2} - \frac{3a^2}{2} \right)} \\
 &= \frac{- \left(\frac{9a^2}{2} - \frac{3a^2}{2} \right)}{\left(\frac{9a^2}{2} - \frac{3a^2}{2} \right)} \quad \because A-B = -(B-A) \\
 &= -1
 \end{aligned}$$

but $\frac{dy}{dx} = \text{slope} = \tan \theta$

$$\text{and } \tan \theta = -1$$

$$\theta = \tan^{-1}(-1)$$

$$\theta = \frac{3\pi}{4} = 135^\circ$$

* Rule 5: Asymptotes

(A) Asymptote parallel to x-axis

$$x^3 + y^3 - 3axy = 0$$

$$\begin{array}{ccc}
 \downarrow & \downarrow & \downarrow \\
 3 & 0 & 1
 \end{array}$$

—————→ powers of x.

To find asymptote parallel to x-axis
let coefficient of highest power of x = 0

$$1 = 0$$

which is not possible

$\therefore$ No asymptote parallel to x-axis.

② Asymptote parallel to Y-axis

$$\text{As } x^3 + y^3 - 3axy = 0$$

$$\begin{array}{ccc} \downarrow & \downarrow & \downarrow \\ 0 & 3 & 1 \end{array}$$

—————→ powers of y.

To find the asymptotes parallel to Y-axis
let

coefficient of highest
power of y

$$1 = 0$$

which is not possible.

∴ No Asymptote parallel to Y-axis.

• ③ Oblique Asymptote (For Implicit functions only)

let $y = mx + c$ be the oblique asymptote.

∴ put $y = mx + c$ in the given eqⁿ.

$$\begin{aligned} x^3 + y^3 &= 3axy \\ x^3 + (mx + c)^3 &= 3ax(mx + c) \end{aligned}$$

$$(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3.$$

$$x^3 + m^3x^3 + 3m^2xc + 3mxc^2 + c^3 = 3amx^2 + 3acx$$

$$(1+m^3)x^3 + (3m^2c)x^2 + (3mc^2)x + c^3 = (3am)x^2 + (3ac)x$$

on Equating coeff. of x^3

$$(1+m^3) = 0$$

$$m^3 = -1$$

$$\therefore \boxed{m = -1}$$

Equating coefficient of x^2 we get,

$$3m^2c = 3am$$

$$\text{but } m = -1$$

$$\therefore (-1)^2c = a(-1)$$

$$\boxed{c = -a}$$

$\therefore$ Equation of oblique asymptote is

$$y = mx + c$$

$$y = -x - a$$

i.e. $\boxed{x + y + a = 0}$

* Rule 6: Region of Absence:-

The given curve

$$x^3 + y^3 = 3axy.$$

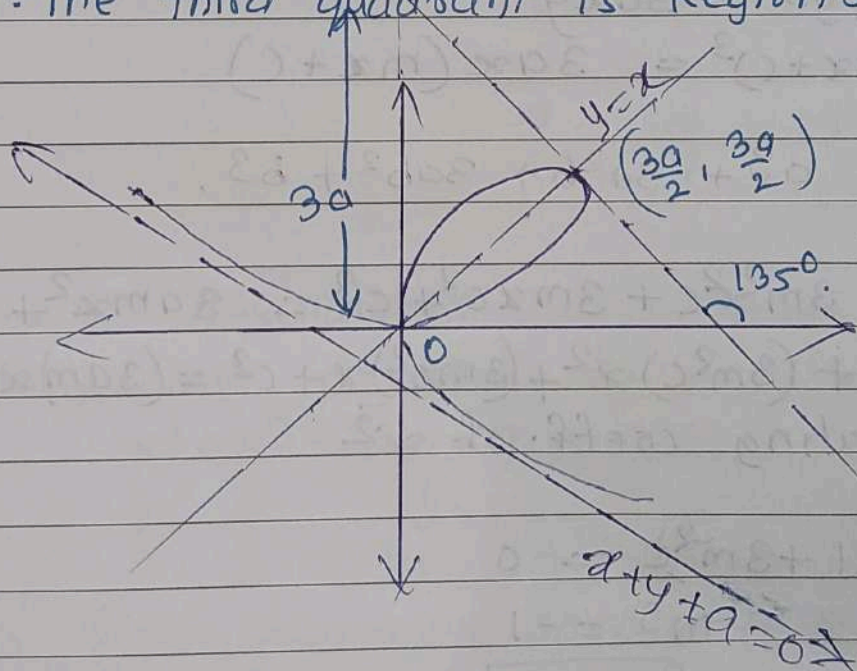
If both x and y become negative then

R.H.S. is +ve and L.H.S. is negative.

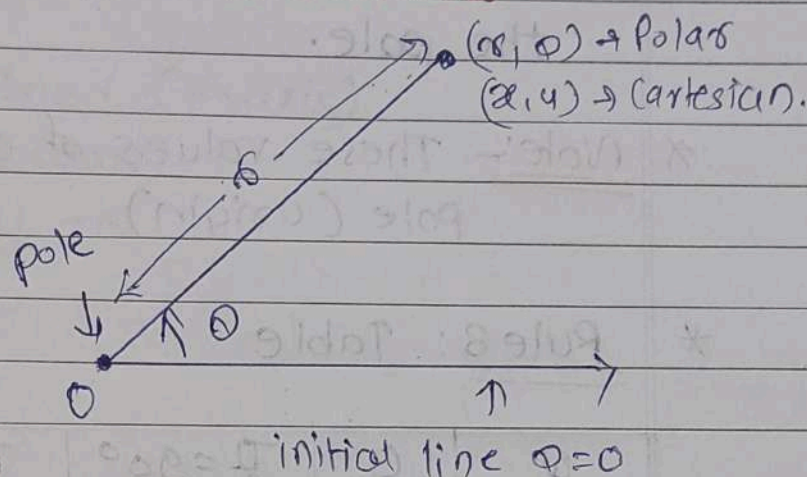
which is not logical.

This means for $x = -ve$ and $y = -ve$ that is third quadrant, there will be no curve.

$\therefore$ The third quadrant is Region of Absence.



* Curves Given by Polar Co-ordinates



* Rule 1: Symmetry

(A) About Initial line (x-axis)

Replace θ by $-\theta$, if the equation of the curve remains unchanged then curve is symmetrical about initial line (x-axis).

(B) About the line $\theta = \frac{\pi}{2}$ (y-axis)

(i) Replace θ by $-\theta$ and r by $-r$ if the equation of curve remains unchanged then the curve is about the line through the pole, perpendicular to initial line (i.e. y-axis).

(ii) The same symmetry also exists if the equation remains unchanged when θ is replaced by $\pi - \theta$.

* Rule 2: Pole

The curve will pass through the pole (origin) if for some value of θ , r becomes $r = 0$.

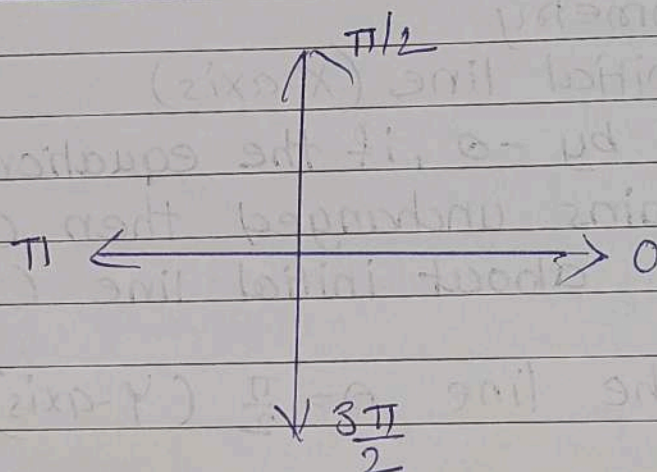
To check it, put $r = 0$ and if we get

valid values of θ , the curve passes through the pole.

* Note:- These values of θ are tangents at pole (origin).

* Rule 3: Table

θ	0	$\frac{\pi}{2} = 90^\circ$	$\pi = 180^\circ$	$270^\circ = \frac{3\pi}{2}$
r				



* Rule 4: Angle between the radius vector and tangent (ϕ) (Tangents at special points)

Use the formula

$$\tan \phi = r \frac{d\theta}{dr} \text{ and find } \phi.$$

Que) Trace the curve $r = a(1 + \cos \theta)$

⇒ Rule 1: Symmetry

(A) About the initial line: (X-axis)

⇒ Given $r = a(1 + \cos \theta)$

Replace θ by $-\theta$

$$\therefore r = a(1 + \cos(-\theta))$$

$$r = a(1 + \cos \theta)$$

$$(\because \cos(-x) = \cos x)$$

$\therefore$ Equation remains unchanged.

$\therefore$ The curve is symmetric about the initial line (X-axis).

(B) About the line $\theta = \frac{\pi}{2}$ (Y-axis).

Given, $r = a(1 + \cos \theta)$

Case (i)

Put $\theta = -\theta$ and $r = -r$ we get

$$-r = a(1 + \cos(-\theta))$$

$$-r = a(1 + \cos \theta)$$

$\therefore$ Equation changes.

$\therefore$ No symmetry about Y-axis.

Case (ii) Replace θ by $\pi - \theta$

$$\therefore r = a(1 + \cos(\pi - \theta))$$

$$r = a(1 - \cos \theta)$$

$$(\because \cos(\pi - \theta) = -\cos \theta)$$

$\therefore$ Equation changes.

$\therefore$ No symmetry about Y-axis.

* Rule 2: Pole

$$\text{Let } r = a(1 + \cos \theta)$$

$$\text{put } r = 0$$

$$0 = a(1 + \cos \theta)$$

$$\therefore 1 + \cos \theta = 0$$

$$\cos \theta = -1$$

$$\theta = \cos^{-1}(-1)$$

$$\theta = \pi, 3\pi, 5\pi, \dots$$

but $\theta = 3\pi, 5\pi \dots$ are values above 360° .

$\therefore$ we will neglect them

hence $\boxed{\theta = \pi}$

$\therefore$ Curve passes through the pole.

As well as these values of θ represents tangents at pole (origin).

* Rule 3: Table.

θ	0	90°	180°	270°
r	$2a$	a	0	a

(i) for $\theta = 0$

$$r = a(1 + \cos \theta)$$

$$r = a(1 + \cos(0))$$

$$r = a(1 + 1)$$

$$\boxed{r = 2a}$$

(ii) for $\theta = 90^\circ$

$$r = a(1 + \cos(90^\circ))$$

$$r = a(1 + 0)$$

$$\boxed{r = a}$$

(iii) for $\theta = 180^\circ$

$$r = a(1 + \cos(180^\circ))$$

$$r = a(1 + (-1))$$

$$r = a(0)$$

$$\boxed{r = 0}$$

(iv) for $\theta = 270^\circ$

$$r = a(1 + \cos(270^\circ))$$

$$r = a(1 + 0)$$

$$\boxed{r = a}$$

* Rule 4: Tangents at special points

Given $r = a(1 + \cos \theta)$

Differentiating w.r.t. θ

$$\frac{dr}{d\theta} = a \times \frac{d}{d\theta} (1 + \cos\theta)$$

$$\frac{dr}{d\theta} = a(0 - \sin\theta)$$

$$\left(\because \frac{d}{dx} \cos x = -\sin x \right)$$

$$\frac{dr}{d\theta} = -a \sin\theta$$

on reciprocal

$$\frac{d\theta}{dr} = \frac{1}{-a \sin\theta}$$

Multiply by r on both sides.

$$r \frac{d\theta}{dr} = \frac{r}{-a \sin\theta}$$

$$\text{but } r = a(1 + \cos\theta)$$

$$\therefore \frac{r d\theta}{dr} = \frac{a(1 + \cos\theta)}{-a \sin\theta}$$

$$\frac{r d\theta}{dr} = -\frac{(1 + \cos\theta)}{\sin\theta}$$

$$\therefore \frac{r d\theta}{dr} = -\frac{2 \cos^2(\theta/2)}{2 \sin(\frac{\theta}{2}) \cos(\frac{\theta}{2})}$$

$$\left(\because 1 + \cos x = 2 \cos^2\left(\frac{x}{2}\right) \right)$$

and

$$\sin 2x = 2 \sin x \cos x$$

$$\frac{r d\theta}{dr} = -\cot\left(\frac{\theta}{2}\right)$$

$$\text{but } \frac{r d\theta}{dr} = \tan \phi$$

$$\therefore \Rightarrow \tan \phi = -\cot\left(\frac{\theta}{2}\right)$$

$$\Rightarrow \tan \phi = \tan\left(\frac{\pi}{2} + \frac{\theta}{2}\right)$$

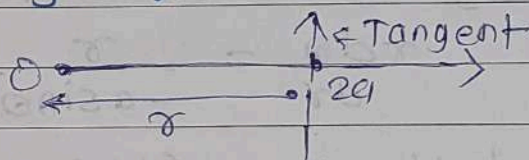
$$\therefore \boxed{\phi = \frac{\pi}{2} + \frac{\theta}{2}}$$

* Table of r , θ and ϕ .

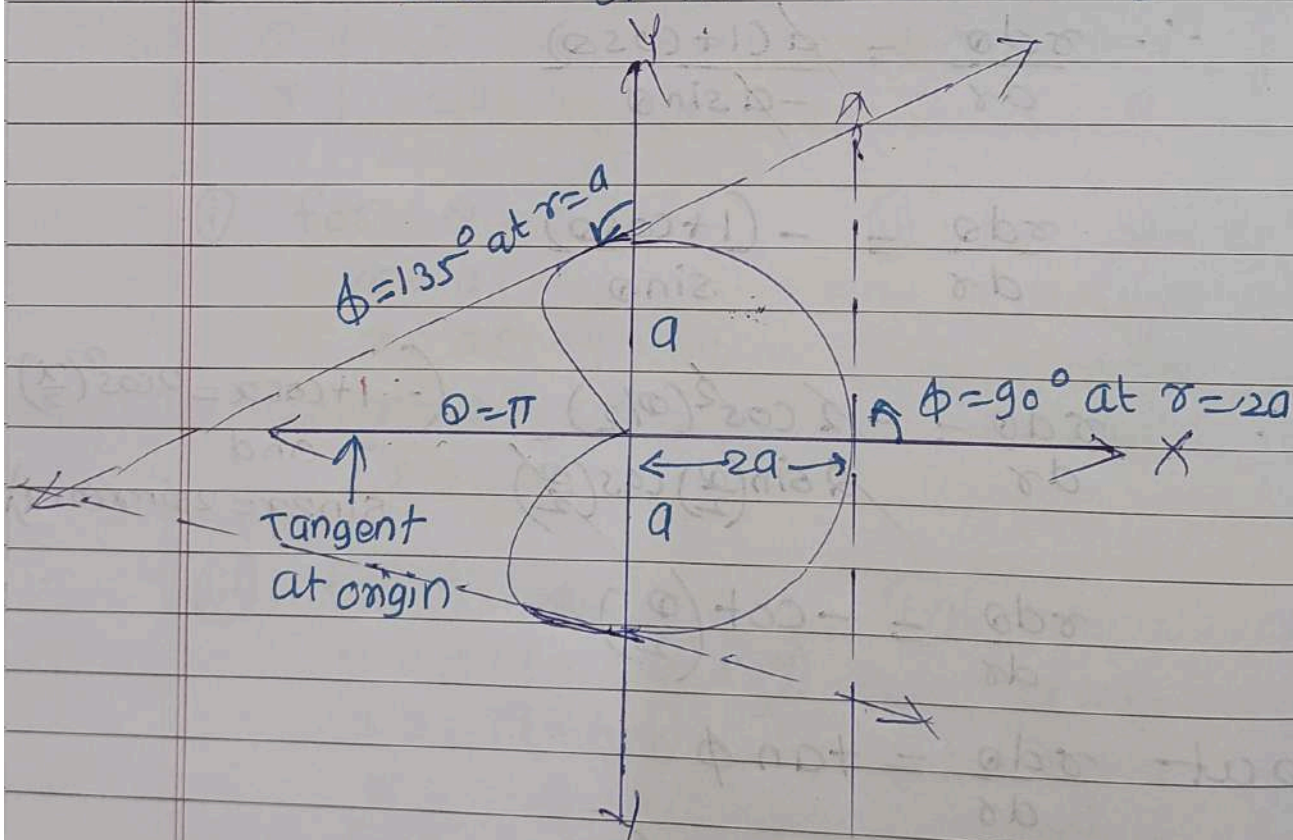
θ	0°	90°	180°	270°
r	$2a$	a	0	a
ϕ	90°	135°	180°	225°

When $\theta = 0^\circ$, we get $\phi = 90^\circ (\frac{\pi}{2})$

i.e. Tangent at 90° with $\theta = 0$ at a distance $r = 2a$.

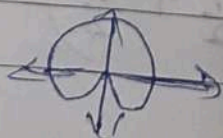


The sketch of the curve is as given below.



Que) Trace the curve $r = a(1 + \sin \theta)$

$\Rightarrow$ symmetric about Y-axis, $\theta = \frac{3\pi}{2}$, $\phi = \frac{\pi}{4} + \frac{\theta}{2}$



* Rose Curves:- Curves given by Polar Coordinates of the type $r = a \sin n\theta$ or $r = a \cos n\theta$.

* Rule 1: Symmetry

(i) Curves given by $r = a \cos n\theta$ are symmetrical about initial line (x-axis)

(ii) Curves given by $r = a \sin n\theta$ are symmetrical about the line perpendicular to initial line (i.e. y-axis)

* Rule 2: For $r = a \sin n\theta$ and $r = a \cos n\theta$, divide each quadrant into 'n' equal parts.

* Rule 3:-

*** VIMP (i) IF n is even then 2n loops

(ii) IF n is odd then n loops

* Rule 4:-

(i) For $r = a \cos(n\theta)$, Draw first loop along $\theta = 0^\circ$.

(ii) For $r = a \sin(n\theta)$, draw first loop along $\theta = \frac{\pi}{2n}$

* Rule 5:- (1) IF n is even, draw loops in every sector continuously

(2) IF n is odd, keep two sectors vacant between the loops.

* Rule 6:- Tangents at origin
put $r=0$ and find different values of θ . Neglect values of θ beyond $360^\circ (2\pi)$

$$\textcircled{1} \cos^{-1}(0) = (2n+1) \frac{\pi}{2}$$

$$\textcircled{2} \sin^{-1}(0) = n\pi \quad \text{where } n = 0, 1, 2, 3, \dots$$

* Rule 7:- Tangents at special points.
Determine the angle ϕ of a tangent using the formula $\phi = r \frac{d\theta}{dr}$

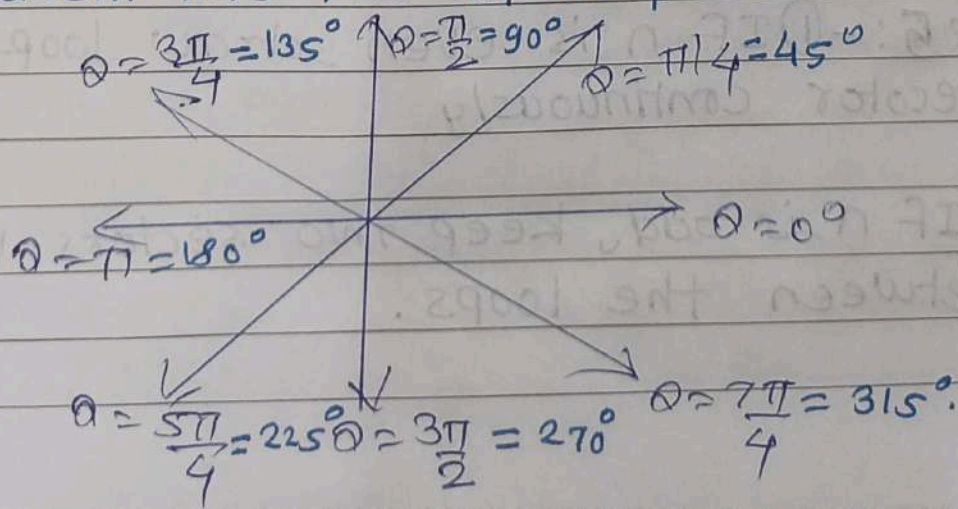
* Rule 8:- Prepare a table of r , θ and ϕ .

Que) Trace the curve $r = a \cos(2\theta)$.
 $\Rightarrow$ Given $r = a \cos 2\theta$
on comparing with $r = a \cos n\theta$
 $\Rightarrow \boxed{n=2} \rightarrow$ even number.

* Rule 1: Symmetry

The curve is symmetric about the initial line.

* Rule 2: Here $n=2$, Therefore divide each quadrant into two equal parts.



* Degree to Radian Conversion *

$$\theta^\circ = \left(\theta \times \frac{\pi}{180} \right)^c$$

* Rule 3:- Here $n=2 \rightarrow$ even no. which implies there will be 4 loops.

* Rule 4:- Here $r = a \cos 2\theta$ therefore 1st loop is along $\theta=0$.

* Rule 5:- Here, $n=2 \rightarrow$ even, Therefore draw loops consecutively in every sector.

* Rule 6:- Tangents at origin we have $r = a \cos 2\theta$
put $r=0$

$$\therefore 0 = a \cos 2\theta$$

$$\cos 2\theta = 0$$

$$2\theta = \cos^{-1}(0)$$

$$2\theta = (2n+1) \frac{\pi}{2} \quad n=0,1,2,3,\dots$$

put the values of n

$$2\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}, \dots$$

(Neglecting values above $360^\circ = 2\pi$)

$$\theta = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}, \dots$$

Gives tangents at origin.

* Rule 7: Tangents at special points.

Given $r = a \cos 2\theta$

Differentiating w.r.t. θ

$$\frac{dr}{d\theta} = -2a \sin(2\theta)$$

on reciprocal

$$\frac{d\theta}{dr} = \frac{1}{-2a \sin(2\theta)}$$

multiply by r on both sides

$$\Rightarrow \therefore r \frac{d\theta}{dr} = \frac{r}{-2a \sin(2\theta)}$$

but $r = a \cos(2\theta)$

$$\Rightarrow \therefore r \frac{d\theta}{dr} = \frac{a \cos(2\theta)}{-2a \sin(2\theta)}$$

$$\Rightarrow \therefore r \frac{d\theta}{dr} = \frac{-1}{2} \cot(2\theta)$$

but $r \frac{d\theta}{dr} = \tan \phi$

$$\Rightarrow \tan \phi = \frac{-1}{2} \cot(2\theta)$$

$$\Rightarrow \tan \phi = \frac{1}{2} \tan\left(\frac{\pi}{2} + 2\theta\right)$$

* Rule 8: - Table of r , θ and ϕ .

from rule two, we have values of ϕ .

$$\theta = 0, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \pi, \frac{5\pi}{4}, \frac{3\pi}{2}, \frac{7\pi}{4}$$

θ	r	ϕ
0°	a	90°
45°	0	0
90°	$-a$	90°
135°	0	0
180°	a	90°
225°	0	0
270°	$-a$	90°
315°	0	0

$$\therefore \tan \phi = \frac{1}{2} \tan \left(\frac{\pi}{2} + 2\theta \right)$$

(i) When $\theta = 0$

$$\tan \phi = \frac{1}{2} \tan \left(\frac{\pi}{2} + 0 \right)$$

$$\tan \phi = \infty$$

$$\phi = \tan^{-1}(\infty)$$

$$\phi = \frac{\pi}{2} = 90^\circ$$

(ii) when $\theta = 45^\circ = \frac{\pi}{4}$

$$\therefore \tan \phi = \frac{1}{2} \tan \left(\frac{\pi}{2} + 2\left(\frac{\pi}{4}\right) \right) = \frac{1}{2} \tan \left(\frac{\pi}{2} + \frac{\pi}{2} \right) = \frac{1}{2} \tan(\pi)$$

$$\tan \phi = 0$$

$$\phi = \tan^{-1}(0)$$

$$\boxed{\phi = 0}$$

... and so on.

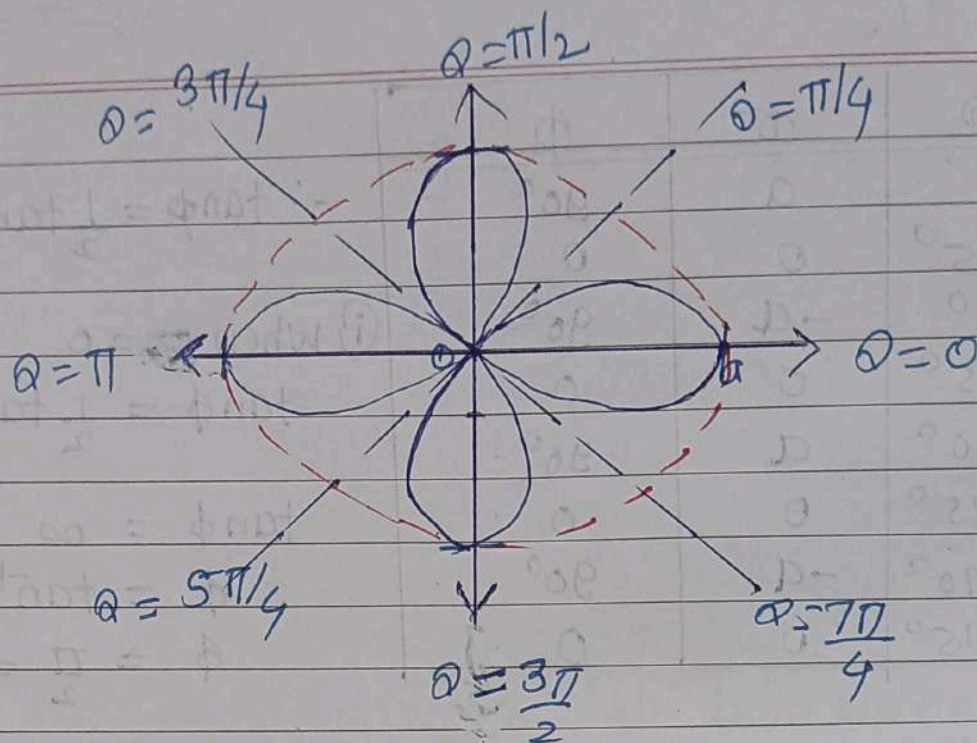
$$\text{For } \theta = 45^\circ \Rightarrow \phi = 0$$

i.e. Tangent and radius vectors are coincident at $r=0$.

(iii) For $\theta = 0 \Rightarrow \phi = 90^\circ$.

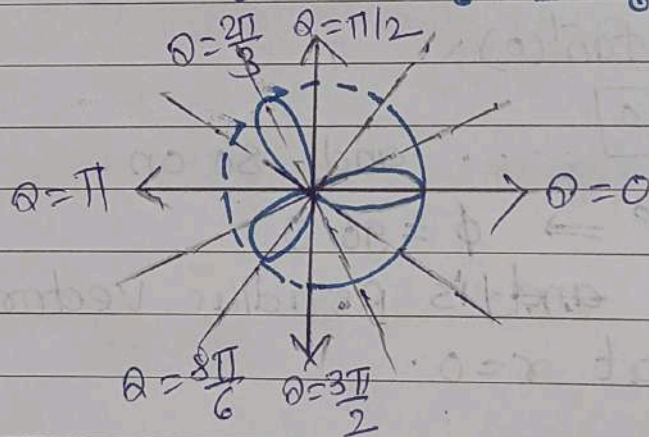
That is tangent is perpendicular to radius vector at a distance $r=a$.

$\therefore$ The sketch of the curve is as



Que Trace the curve $r = a \cos 3\theta$.
 H.W. $\Rightarrow \tan \phi = \frac{1}{3} \tan \left(\frac{\pi}{2} + 3\theta \right)$, 3 loops

Tangents at origin, $\theta = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{3\pi}{2}, \frac{11\pi}{6}$.



* Curves given by Parametric Equations:-
consider the parametric equation given by
 $x = f(t)$ and $y = g(t)$

* Rule 1: Symmetry

(i) IF $x = f(t)$ is an even function and $y = g(t)$ is an odd function then curve is symmetrical about X-axis

$$\therefore \text{Even} \Rightarrow f(-x) = f(x)$$

$$\text{odd} \Rightarrow f(-x) = -f(x)$$

(ii) IF $x = f(t)$ is an odd function and $y = g(t)$ is an even function then curve is symmetrical about Y-axis.

(iii) IF $x = f(t)$ and $y = g(t)$ both are odd, then curve is symmetrical in opposite quadrants.

(iv) For values of t and $-t$, x remains unchanged but y has equal and opposite values, then curve is symmetrical about X-axis

(v) Replace t by $(\pi - t)$, if y has same value and x has equal and opposite value, then curve is symmetrical about Y-axis.

* Rule 2: Origin

IF for some values of t , both x and y are zero then curve passes through origin.

* steps *

step ① :- put $x=0$ and find t

step ② :- Put this value of t in y .

step ③ IF we get $y=0$ for the same value of t , then curve passes through the origin.

* Rule 3 :- Tangents at special points.

find $\frac{dy}{dx}$ by using

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$$

* Rule 4: Prepare a table of $t, x, y, \frac{dy}{dx}$.

Que) Trace the curve $x^{2/3} + y^{2/3} = a^{2/3}$.

or
Trace the curve $\left(\frac{x}{a}\right)^{2/3} + \left(\frac{y}{a}\right)^{2/3} = 1$.

⇒ This curve is known as Asteroïd. Also known as star shaped curve.

The parametric equations of the curve are given by

$$x = a \cos^3 t \quad \text{and} \quad y = a \sin^3 t.$$

* Rule 1: Symmetry.

$$\text{Let } x = f(t) = a \cos^3 t$$

put $t = -t$ we get

$$f(-t) = a [\cos(-t)]^3$$

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$$f(t) = a(\cos t)^3 \quad (\because \cos(-\theta) = \cos \theta)$$

$$f(-t) = a\cos^3 t$$

$$f(-t) = f(t)$$

$\therefore x = f(t)$ is an even function.

Now let $y = g(t) = a\sin^3 t$

put $t = -t$ we get

$$g(-t) = a[\sin(-t)]^3$$

$$g(-t) = -a\sin^3 t \quad (\because \sin(-\theta) = -\sin \theta)$$

$$g(-t) = -g(t)$$

Hence y is an odd function.

As x is even and y is odd, Curve is symmetrical about x -axis.

(ii) If we do not get symmetry about y axis in Rule 1 then we can check as follows.

As $x = a\cos^3 t$

Replace t by $(\pi - t)$

$$\therefore x = a[\cos(\pi - t)]^3$$

$$x = a(-\cos t)^3$$

$$x = -a\cos^3 t$$

$\therefore x$ has opposite value.

Also $y = a\sin^3 t$

Replace t by $(\pi - t)$.

$$y = a[\sin(\pi - t)]^3$$

$$\sin(\pi - \theta) = \sin \theta$$

$$\cos(\pi - \theta) = -\cos \theta$$

$$y = a (\sin t)^3$$

$$y = a \sin^3 t$$

y has same value.

After replacing t by $(\pi - t)$,
 x has opposite value and y has same value. Hence curve is symmetrical about y -axis.

* Rule 2: Origin

We have $x = a \cos^3 t$

put $x = 0$

$$0 = a \cos^3 t$$

$$\cos^3 t = 0$$

$$\cos t = 0$$

$$t = \cos^{-1}(0)$$

$$t = \frac{\pi}{2} = 90^\circ$$

Now substitute $t = 90^\circ$ in $y = a \sin^3 t$

$$\therefore y = a [\sin(90^\circ)]^3$$

$$y = a (1)^3$$

$$\boxed{y = a}$$

$$(\because \sin(90^\circ) = 1)$$

For $t = \frac{\pi}{2}$, and $x = 0$, $y = a$ i.e. $(0, a)$.

Curve does not pass through origin...

* Rule 3: Find $\frac{dy}{dx}$,

$$\text{As } y = a \sin^3 t$$

diff. w.r.t. 't'.

$$\frac{dy}{dt} = a \cdot 3 \sin^2 t \cdot \cos t = 3a \sin^2 t \cos t.$$

Now $x = a \cos 3t$

diff. w.r.t. t

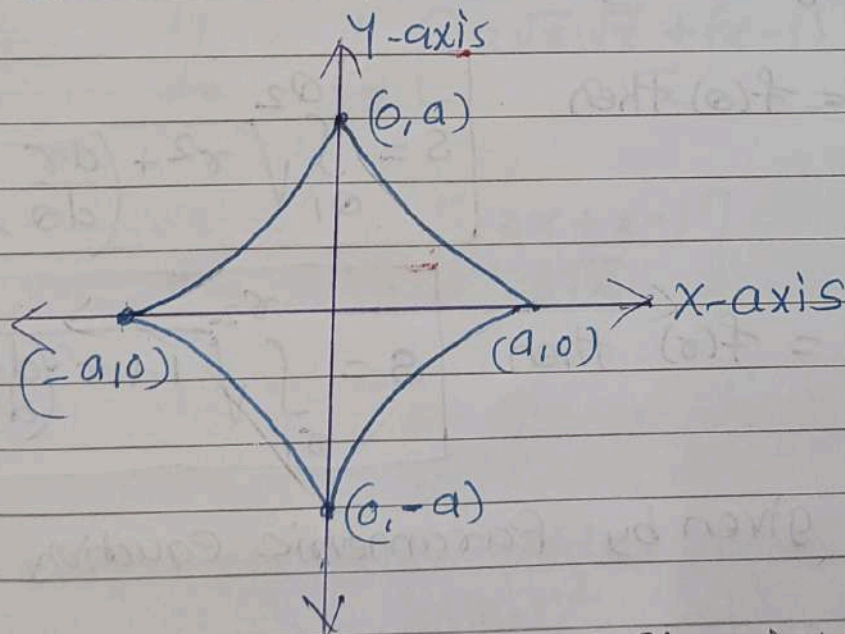
$$\frac{dx}{dt} = a \cdot 3 \cos^2 t (-\sin t) = -3a \cos^2 t \sin t.$$

$$\therefore \frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{3a \sin^2 t \cos t}{-3a \cos^2 t \sin t} = -\tan t$$

* Rule 4: Prepare a table for t, x, y, $\frac{dy}{dx}$.

t	0	$\frac{\pi}{2} = 90^\circ$	$\pi = 180^\circ$	$\frac{3\pi}{2} = 270^\circ$	$2\pi = 360^\circ$
x	a	0	-a	0	a
y	0	a	0	-a	0
dy/dx	0	$-\infty$	0	∞	0

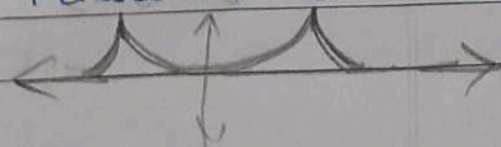
The sketch of the curve is as follows.



Que. Trace the cycloid, $x = a(t + \sin t)$, $y = a(1 - \cos t)$

(i) Symmetric about Y-axis, (ii) Passes through origin.

(iii) $\frac{dy}{dx} = \tan\left(\frac{t}{2}\right)$.



Unit 4, chapter-6

'Rectification of curves'

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* Reference book -

Higher Engineering Mathematics By B.S. Grewal
(Khanna Publications Delhi)

* **Rectification of curves:-** the process of finding the length of ~~curve~~ an arc of a curve is called as Rectification of curve.

* Curves given by Cartesian co-ordinates.

① IF $y = f(x)$ then
$$s = \int_{x_1}^{x_2} \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

② IF $x = g(y)$ then
$$s = \int_{y_1}^{y_2} \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

* Curves given by Polar co-ordinates.

③ IF $r = f(\theta)$ then
$$s = \int_{\theta_1}^{\theta_2} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$

④ IF $\theta = f(r)$ then
$$s = \int_{r_1}^{r_2} \sqrt{1 + r^2 \left(\frac{d\theta}{dr}\right)^2} dr$$

* Curves given by Parametric Equations.

⑤ IF $x = f(t)$ and
 $y = g(t)$

$$s = \int_{t_1}^{t_2} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

Que) Find the perimeter of the loop of the curve

$$3y^2 = x(x-1)^2$$

⇒ Step ① find $\sqrt{1 + \left(\frac{dy}{dx}\right)^2}$

Given, $3y^2 = x(x-1)^2$

$$y^2 = \frac{x}{3}(x-1)^2$$

$$y = \frac{\sqrt{x}}{\sqrt{3}}(x-1)$$

Differentiate w.r.t. x

$$\frac{dy}{dx} = \frac{1}{\sqrt{3}} \cdot \frac{d}{dx} [\sqrt{x}(x-1)]$$

$$\frac{dy}{dx} = \frac{1}{\sqrt{3}} \left[\sqrt{x}(1-0) + (x-1) \cdot \frac{1}{2\sqrt{x}} \right]$$

$$\frac{dy}{dx} = \frac{1}{\sqrt{3}} \left[\sqrt{x} + \frac{(x-1)}{2\sqrt{x}} \right]$$

$$\frac{dy}{dx} = \frac{1}{\sqrt{3}} \left[\frac{2\sqrt{x} \cdot \sqrt{x} + (x-1)}{2\sqrt{x}} \right]$$

$$\frac{dy}{dx} = \frac{1}{\sqrt{3}} \left[\frac{2x + x - 1}{2\sqrt{x}} \right]$$

$$\frac{dy}{dx} = \frac{3x-1}{2\sqrt{3x}}$$

on squaring both sides

$$\left(\frac{dy}{dx}\right)^2 = \frac{(3x-1)^2}{(4)(3x)}$$

$$\therefore \left(\frac{dy}{dx}\right)^2 = \frac{9x^2 - 6x + 1}{12x}$$

Add 1 on both sides

$$1 + \left(\frac{dy}{dx}\right)^2 = 1 + \frac{9x^2 - 6x + 1}{12x}$$

$$\therefore 1 + \left(\frac{dy}{dx}\right)^2 = \frac{12x + 9x^2 - 6x + 1}{12x}$$

$$1 + \left(\frac{dy}{dx}\right)^2 = \frac{9x^2 + 6x + 1}{12x}$$

$$1 + \left(\frac{dy}{dx}\right)^2 = \frac{(3x+1)^2}{12x}$$

Taking square root on both side

$$\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \sqrt{\frac{(3x+1)^2}{12x}}$$

$$\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \frac{(3x+1)}{\sqrt{12} \cdot \sqrt{x}}$$

step ② find limits of integration (x_1 and x_2)

$$\text{Given } 3y^2 = x(x-1)^2$$

$$\text{put } y=0$$

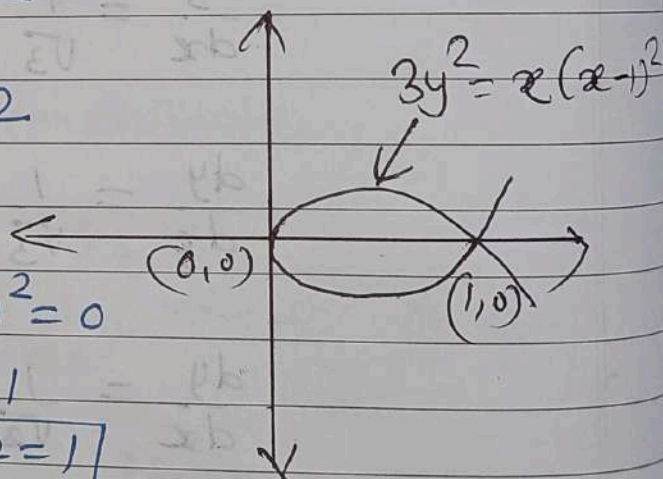
$$0 = x(x-1)^2$$

$$x(x-1)^2 = 0$$

$$\therefore x=0 \quad \text{or} \quad (x-1)^2=0$$

$$x=0 \quad \text{or} \quad x=1$$

$$\therefore \boxed{x_1=0} \quad \text{and} \quad \boxed{x_2=1}$$



step ③ find perimeter 'S'.

$$S = \int_{x_1}^{x_2} \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$S = \int_0^1 \frac{(3x+1)}{\sqrt{12} \cdot \sqrt{x}} dx.$$

$$\therefore S = \frac{1}{\sqrt{12}} \int_0^1 \left(\frac{3x}{x^{1/2}} + \frac{1}{x^{1/2}} \right) dx$$

$$S = \frac{1}{\sqrt{12}} \int_0^1 \left(3x^{(1-\frac{1}{2})} + x^{-\frac{1}{2}} \right) dx$$

$$\therefore \frac{a^m}{a^n} = a^{m-n}$$

$$\therefore S = \frac{1}{\sqrt{12}} \left[\int_0^1 3x^{1/2} + x^{-1/2} \right] dx$$

$$S = \frac{1}{\sqrt{12}} \left[\frac{3x^{3/2}}{(3/2)} + \frac{x^{1/2}}{(1/2)} \right]_0^1 \quad \because \int x^n dx = \frac{x^{n+1}}{n+1}$$

$$S = \frac{1}{\sqrt{12}} \left[2x^{3/2} + 2x^{1/2} \right]_0^1$$

$$S = \frac{1}{\sqrt{12}} \left[(2(1)^{3/2} + 2(1)^{1/2}) - (0+0) \right]$$

$$S = \frac{4}{\sqrt{12}}$$

$$\text{But for symmetry } S = 2 \times \frac{4}{\sqrt{12}} = \frac{8}{\sqrt{12}} = \frac{4 \times 2}{2\sqrt{3}} = \frac{4}{\sqrt{3}}$$

$\therefore$ The perimeter of the loop of the curve is $\frac{4}{\sqrt{3}}$ unit.

Que) Find the length of the upper arc of one loop of Lemniscate $r^2 = a^2 \cos(2\theta)$

$\Rightarrow$ Step ① find $\sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}$.

$$\text{Given } r^2 = a^2 \cos(2\theta)$$

differentiate w.r.t θ , we get

$$2r \frac{dr}{d\theta} = a^2 (-\sin(2\theta)) \times 2$$

$$r \frac{dr}{d\theta} = -a^2 \sin(2\theta)$$

$$\therefore \frac{dr}{d\theta} = \frac{-a^2 \sin(2\theta)}{r}$$

Squaring on both sides

$$\Rightarrow \left(\frac{dr}{d\theta} \right)^2 = \left(\frac{-a^2 \sin(2\theta)}{r} \right)^2$$

$$\Rightarrow \left(\frac{dr}{d\theta} \right)^2 = \frac{a^4 \sin^2(2\theta)}{r^2}$$

$$\text{but } r^2 = a^2 \cos(2\theta).$$

$$\therefore \left(\frac{dr}{d\theta} \right)^2 = \frac{a^4 \sin^2(2\theta)}{a^2 \cos(2\theta)}$$

$$\left(\frac{dr}{d\theta} \right)^2 = \frac{a^2 \sin^2(2\theta)}{\cos(2\theta)}$$

Add r^2 on both sides.

$$\therefore r^2 + \left(\frac{dr}{d\theta} \right)^2 = r^2 + \frac{a^2 \sin^2(2\theta)}{\cos(2\theta)}$$

$$\text{but } r^2 = a^2 \cos(2\theta)$$

$$\therefore r^2 + \left(\frac{dr}{d\theta} \right)^2 = a^2 \cos(2\theta) + \frac{a^2 \sin^2(2\theta)}{\cos(2\theta)}$$

$$= \frac{a^2 \cos^2(2\theta)}{\cos(2\theta)} + \frac{a^2 \sin^2(2\theta)}{\cos(2\theta)}$$

$$= \frac{a^2 [\cos^2(2\theta) + \sin^2(2\theta)]}{\cos(2\theta)}$$

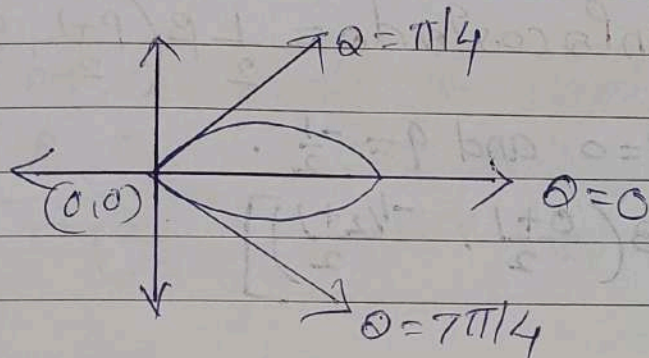
$$r^2 + \left(\frac{dr}{d\theta} \right)^2 = \frac{a^2 (1)}{\cos(2\theta)}$$

Taking square root on both side.

$$\sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} = \frac{a}{\sqrt{\cos 2\theta}}$$

Step ② : To find limits of integration.

The curve $r^2 = a^2 \cos 2\theta$ is given by



To find the length of an upper arc.

$$\theta_1 = 0 \text{ and } \theta_2 = \frac{\pi}{4}$$

Step ③ To find length,

Formula,
$$s = \int_{\theta_1}^{\theta_2} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$

$$\therefore s = \int_0^{\pi/4} \frac{a}{\sqrt{\cos 2\theta}} d\theta$$

$$\Rightarrow s = \int_0^{\pi/4} a (\cos 2\theta)^{-1/2} d\theta \quad \text{--- (1)}$$

put $2\theta = t$

$$\theta = \frac{t}{2}$$

differentiate w.r.t. θ .

$$1 = \frac{1}{2} \frac{dt}{d\theta}$$

$$\boxed{d\theta = \frac{1}{2} dt}$$

As $\theta = 0 \Rightarrow t = 2(0) = 0$

$\theta = \pi/4 \Rightarrow t = 2(\pi/4) = \frac{\pi}{2}$

$$\therefore t=0 \text{ and } t=\pi/2$$

$$\therefore S = a \int_0^{\pi/2} (\cos t)^{-1/2} \frac{dt}{2}$$

$$S = \frac{a}{2} \int_0^{\pi/2} (\sin t)^0 (\cos t)^{-1/2} dt$$

Beta Function, $\int_0^{\pi/2} \sin^p \theta \cos^q \theta d\theta = \frac{1}{2} \beta\left(\frac{p+1}{2}, \frac{q+1}{2}\right)$

Here $p=0$ and $q=-\frac{1}{2}$.

$$\therefore S = \frac{a}{2} \left[\frac{1}{2} \beta\left(\frac{0+1}{2}, \frac{-1/2+1}{2}\right) \right]$$

$$S = \frac{a}{2} \left[\frac{1}{2} \beta\left(\frac{1}{2}, \frac{1}{4}\right) \right]$$

$$S = \frac{a}{4} \left[\frac{\sqrt{\frac{1}{2}} \cdot \sqrt{\frac{1}{4}}}{\sqrt{\frac{1}{2} + \frac{1}{4}}} \right] \quad \left(\because \beta(m, n) = \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)} \right)$$

$$S = \frac{a}{4} \left[\frac{\sqrt{\pi} \cdot \sqrt{1/4}}{\sqrt{3/4}} \right] \quad \because \sqrt{\frac{1}{2}} = \sqrt{\pi}$$

$$S = \frac{a}{4} \frac{\sqrt{\pi} \cdot \sqrt{1/4}}{\sqrt{3/4}}$$

By symmetry $S = 2 \times \frac{a}{4} \frac{\sqrt{\pi} \cdot \sqrt{1/4}}{\sqrt{3/4}} = \frac{a}{2} \frac{\sqrt{\pi} \sqrt{1/4}}{\sqrt{3/4}}$

$\therefore$ Hence required length of curve $= S = \frac{a}{2} \frac{\sqrt{\pi} \sqrt{1/4}}{\sqrt{3/4}}$

✱

- ①

⇒

Hint:

- ②

$\Rightarrow$

due)

$$x^{2/3} + y^{2/3} = a^{2/3}.$$

→

$$x = a \cos^3 \theta, \quad y = a \sin^3 \theta.$$

Step ① find $\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$

We have $x = a \cos^3 \theta$

diff w. s.t. @

$$\Rightarrow \frac{dr}{d\theta} = -3a \cos 2\theta \cdot \sin \theta$$

on squaring both sides

$$\Rightarrow \left(\frac{dx}{da} \right)^2 = ga^2 \cos^4 \theta \sin^2 \theta \dots (1)$$

Also $y = a \sin^3 \theta$

diff. w.o.t. @

$$\frac{dy}{d\theta} = 3a \sin^2 \theta \cos \theta$$

Squaring on both sides

$$\left(\frac{dy}{d\theta}\right)^2 = 9a^2 \sin^4 \theta \cos^2 \theta \quad \text{--- ②}$$

Add eqⁿs ① and ②

$$\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2 = 9a^2 \cos^4 \theta \sin^2 \theta + 9a^2 \sin^4 \theta \cos^2 \theta$$

$$= 9a^2 \cos^2 \theta \sin^2 \theta (\cos^2 \theta + \sin^2 \theta)$$

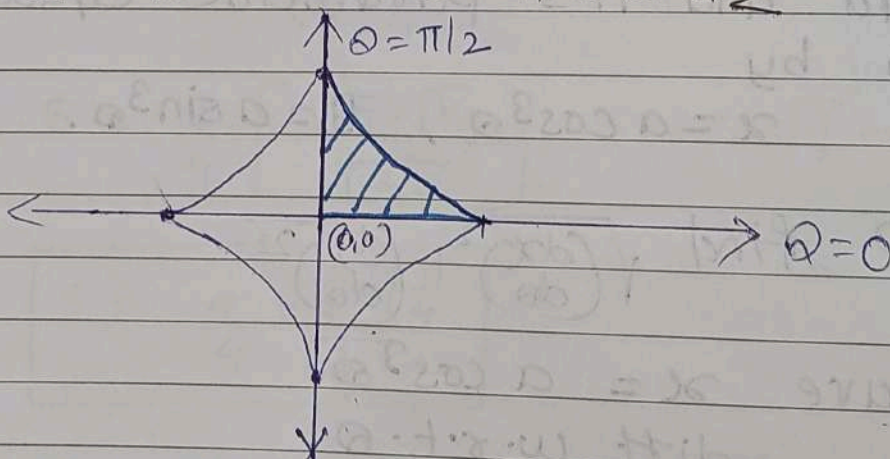
$$= 9a^2 \cos^2 \theta \sin^2 \theta (1)$$

$$\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2 = 9a^2 \cos^2 \theta \sin^2 \theta$$

Taking square root on b.s.

$$\sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} = 3a \cos \theta \sin \theta$$

Step ② find limits θ_1 and θ_2



By symmetry, consider $\theta_1 = 0$ and $\theta_2 = \pi/2$.

Step ③ Calculate length

$$S = \int_{\theta_1}^{\theta_2} \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} d\theta$$

$$S = \int_0^{\pi/2} 3a \cos \theta \sin \theta d\theta$$

$$\therefore S = 3a \int_0^{\pi/2} \sin \theta \cdot \cos \theta d\theta.$$

$$m=1, n=1$$

By Reduction Formula :

$$S = 3a \left[\frac{(1)(1) \cdot 1}{2} \right] \int_0^{\pi/2} \sin^m \theta \cos^n \theta d\theta = \frac{(m-1)(m-3)\dots}{(n-1)(n-3)\dots} \frac{(m+n-2)}{(m+n)}$$

$$S = \frac{3a}{2}$$

$\therefore$ But by Symmetry, total length is

$$S = 4 \times \frac{3a}{2} = 6a.$$

$\therefore$ Complete arc length of given asteroïd is $6a$.

H.W. (1) Find the arc length of the cycloid $x = a(\theta + \sin \theta)$
 Que $y = a(1 - \cos \theta)$ from one cusp to another cusp.

$$\Rightarrow \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} = 2a \cos\left(\frac{\theta}{2}\right), \quad \theta_1 = 0, \theta_2 = \pi, \quad S = 4 \times \frac{2a}{2} = 8a.$$